

Optimizing Heat Source Arrangement for 3D ICs with irregular structures using machine learning methods

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Abstract—Three-dimensional integrated circuits (3D ICs) with irregular structures are recognized as a promising direction for the future development of heterogeneously integrated devices. Due to its high level of integration, 3D ICs encompass an increasing number of stacked chips and heat sources per chip layer, thereby rendering the internal temperature of 3D ICs susceptible to thermal interactions both between and within the layers. Consequently, the optimization of heat source placement on each layer assumes critical importance in order to minimize the maximum temperature and temperature gradient. In this paper, we propose a methodology to optimize the arrangement of heat sources in a three-layer irregular structure 3D ICs. Our approach efficiently combines COMSOL and MATLAB, utilizing a large number of initial samples that are subsequently streamlined through constrained sampling. By assigning diverse weights, we convert the multi-objective optimization problem into a single-objective optimization. Ultimately, we leverage a combination of artificial neural networks and genetic algorithms to achieve the optimal arrangement of six heat sources. To validate the accuracy of our proposed methodology, we conduct a comparative analysis against results obtained from finite element simulation. The comparison highlights the capability of our methodology to rapidly and precisely determine the optimal arrangement of heat source positions, with a maximum temperature deviation of merely 0.24%.

Keywords—irregular structure, 3D ICs, thermal optimization, heat source position

I. INTRODUCTION

The increasing demand for improved integration has generated significant interest in the field of three-dimensional integrated circuits (3D ICs). By utilizing vertical interconnects known as through-silicon vias (TSVs), 3D stacked architecture designs offer notable advantages such as shorter interconnect lengths, higher integration levels, and enhanced performance [1][2]. However, the associated heat-related challenges have become more pronounced [3][4]. In the context of the manufacturing and packaging process, the design of 3D ICs often necessitates irregularly structured chips [5]. Consequently, irregular structure 3D ICs exhibit a less efficient heat transfer path compared to the regular

structure 3D ICs, leading to more severe heat dissipation issues [6].

With the increasing number of chip layers and heat sources, the optimization of heat source locations in irregularly structured 3D ICs is becoming increasingly critical. A careful and accurate choice is vital to avoid undesirable consequences such as excessive chip temperatures or steep temperature gradients across layers, both of which can significantly impact chip performance and reliability [7][8]. Moreover, considering the computational costs associated with finite element analysis, it becomes impractical to exhaustively evaluate all possible heat source locations. Consequently, harnessing the power of machine learning methods becomes imperative for constructing proxy models that enable predictive optimization. Machine learning techniques excel in constructing nonlinear regression models by judiciously selecting representative samples for training, thereby facilitating high optimization accuracy [9][10].

In this study, a machine learning approach was utilized to optimize the arrangement of heat sources in three-layer 3D ICs with irregular structures. The primary objective of optimization was to strike a balance between minimizing the maximum temperature across the entire structure and improving temperature uniformity. Specifically, the focus was on achieving an optimal heat source arrangement that effectively addressed these thermal performance metrics.

II. MODELS AND METHODS

A. Irregular structure 3D ICs model

Figure 1 presents the detailed model of an irregularly structured 3D IC, which was initially introduced in previous work [11]. The model consists of three layers of chips with varying sizes. Each chip layer comprises an overlapping area containing TSVs, while the surrounding area does not include TSVs. The size specifications of each chip structure can be found in TABLE I, while the corresponding thermal conductivity values are provided in TABLE II.

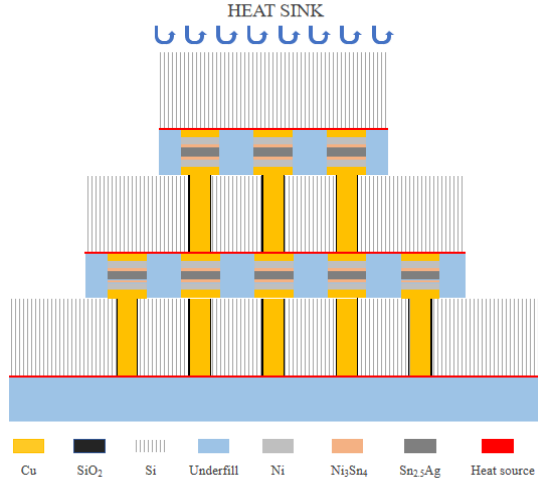


Fig. 1. Detailed model of 3D ICs with irregular structure

TABLE I. THERMAL CONDUCTIVITY OF EACH MATERIAL IN THE MODEL

Material	Conductivity ($\text{Wm}^{-1}\text{K}^{-1}$)
Copper	390
Si	150
SiO ₂	1.5
Ni	60.7
Ni ₃ Sn ₄	19.6
Sn _{2.5} Ag	57
Underfill	0.2

TABLE II. PARAMETERS OF EACH STRUCTURE IN THE MODEL

Structure	Value (μm)
First layer chip (from bottom to top)	280×280×50
Second layer chip	200×200×50
Third layer chip	120×120×50
Diameter of Cu in TSV	16
Diameter of TSV containing SiO ₂ layer	18
Height of micro bumps	23
Diameter of micro bumps	20
Thickness of Ni	3
Thickness of Ni ₃ Sn ₄	1.5
Thickness of Sn _{2.5} Ag	4
Thickness of Cu in the micro-bump	5

To enhance the efficiency of finite element calculations, TSVs and microbumps in the detailed model were equivalently represented as anisotropic thermal blocks and cylinders, respectively. The equivalent model was shown in Fig. 2, and the specific solution process of the equivalent method can be found in our previous works [11][12], where it has been extensively discussed and described.

To expedite and improve the accuracy of solving the steady-state temperature field of the model, the following common assumptions were applied [13]:

(1) Isotropic Layer Materials: Different materials in each layer were assumed to be isotropic, neglecting the contact thermal resistance between adjacent layers.

(2) Isothermal Boundary: An isothermal boundary condition, such as a temperature of 20°C, was imposed on the top surface of the topmost chip. Additionally, all outer

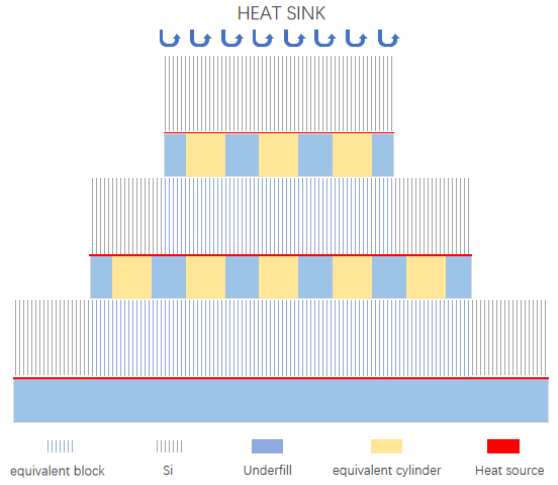


Fig. 2. Equivalent model of 3D ICs with irregular structure surfaces in the model, except for the hot and cold sides, were set as thermal insulation boundaries.

The steady-state temperature field of the model was solved using COMSOL Multiphysics 5.5 software simulation, which facilitated efficient and accurate computation of the thermal distribution within the 3D ICs structure.

B. Heat source location indication

Figure 3 depicts a schematic diagram of the arrangement parameters of two heat sources on each layer of the chip in a three-layer irregular structure 3D ICs. In this representation, each layer of the chip is divided into smaller blocks, totaling 9, 25, and 49 pieces, respectively. Each block measures $40 \mu\text{m} \times 40 \mu\text{m}$ in size. By employing this partitioning approach, machine learning methods can be employed to determine the optimal position arrangement of the two heat sources on each layer of the chip.

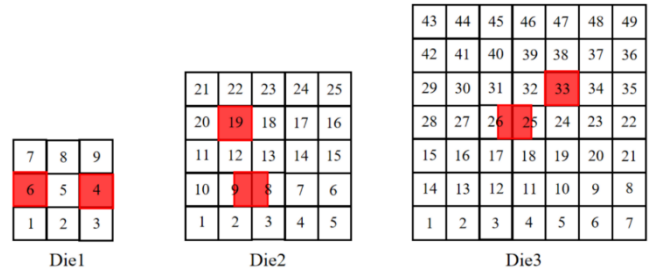


Fig. 3. Schematic diagram of the arrangement parameters of two heat sources on each layer of the chip

C. Sample Generation and Simplification

Figure 4 illustrates the architecture of the co-simulation between COMSOL and MATLAB. In the COMSOL simulation software, the heat source location, heat source power, maximum and average temperature for subsequent analysis can be displayed in the interface. In the model, the goal is to obtain the maximum temperature and the average temperature of the irregularly structured 3D ICs for different heat source positions. The data is then saved and exported as code.

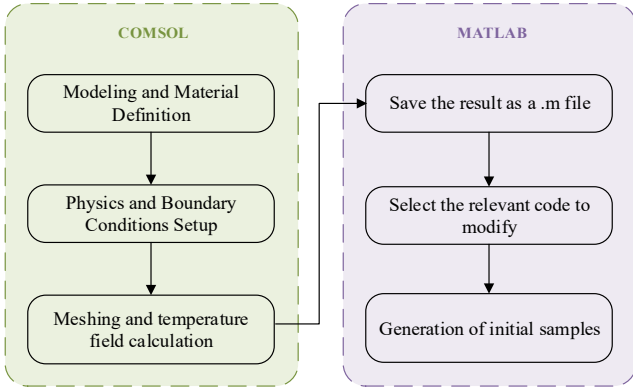


Fig. 4. COMSOL Multiphysics 5.5 with MATLAB co-simulation architecture diagram

In MATLAB, the objective is to locate the heat source, determine the power level, and solve the corresponding maximum and average temperatures using the exported code. These values are combined with the generated code to calculate the temperatures at each small piece of the chip where the heat source is located. The code is meticulously debugged to ensure its accuracy, and then utilized to generate input and output data sets within the MATLAB workspace.

It is important to note that the initial heat source location data generated in this process can be extensive and time-consuming, potentially encompassing tens of millions of data sets. To address the challenge of managing a large initial sample of heat source locations while maintaining representative characteristics, self-learning and adaptive techniques are utilized within spatial constraints. The objective is to streamline the initial sample while ensuring its validity and relevance. This process involves three distinct stages of sampling, as illustrated in Fig. 5. These stages allow for a systematic and controlled approach to generating a representative subset of heat source locations, balancing computational efficiency and accuracy in the optimization process.

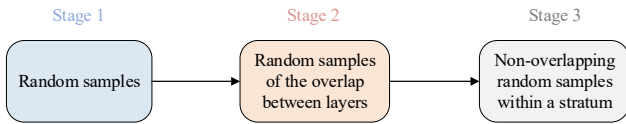


Fig. 5. Sampling phase diagram

D. Artificial neural networks

To obtain more accurate regression functions as an alternative model for optimization, artificial neural networks, which serve as nonlinear regression tools, are employed. These neural networks are developed in MATLAB and trained using the constrained sampled data to solve the fitness function. The three-layer 3D ICs in the model consists of two heat sources in each layer, resulting in a total of six heat sources. Therefore, the input layer of the neural network comprises six nodes to represent the heat source locations. On the other hand, the output layer represents the maximum temperature and average temperature, which do not conflict with each other. Hence, the output layer has only one node. For training the neural network, 70% of the sample size is allocated for training, 15% for testing, and the remaining 15% for validation. In general, to solve most complex problems, a single-hidden layer neural network is sufficient. In the selection of the number of hidden layer nodes, various numbers of nodes are tried during the training process. The

number of nodes is gradually increased from a smaller value, and the error size of the trained results for different node numbers is compared. The aim is to identify a relatively suitable number of nodes with a smaller error. It is ideal to choose a smaller number of nodes when the error is not significant. Additionally, the number of hidden layer nodes selected should be as small as possible, preferably less than the number of training samples (N). The constructed neural network model is shown in Fig. 6.

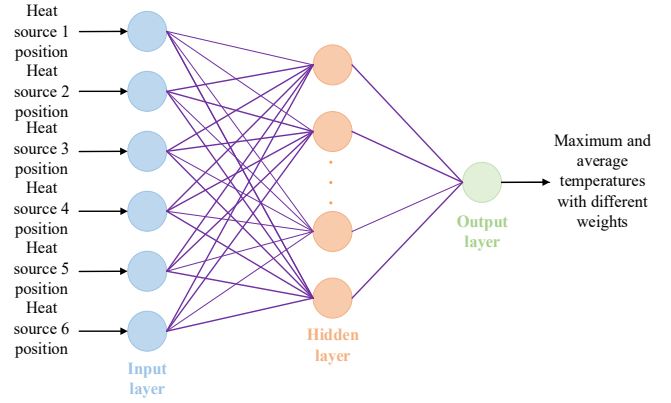


Fig. 6. Models for neural networks

E. Genetic algorithm

The Genetic Algorithm (GA) is a powerful numerical optimization technique inspired by evolutionary principles, such as natural selection. It is widely employed to determine the global optimal solution within a feasible domain. In the GA-ANN optimization process, a population size of 50 is utilized. The selection function employed is Stochunif random uniform selection, which randomly selects individuals based on their fitness. The mutation probability is set to 0.01, indicating the likelihood of genetic mutation occurring in an individual. The crossover probability, set to 0.8, determines the probability of exchanging genetic material between two individuals. These operations are repeated until the optimal individuals in the final population are decoded. The GA-ANN optimization process concludes by obtaining the optimal solution that satisfies the given requirements. The GA-ANN optimization flowchart is depicted in Fig. 7.

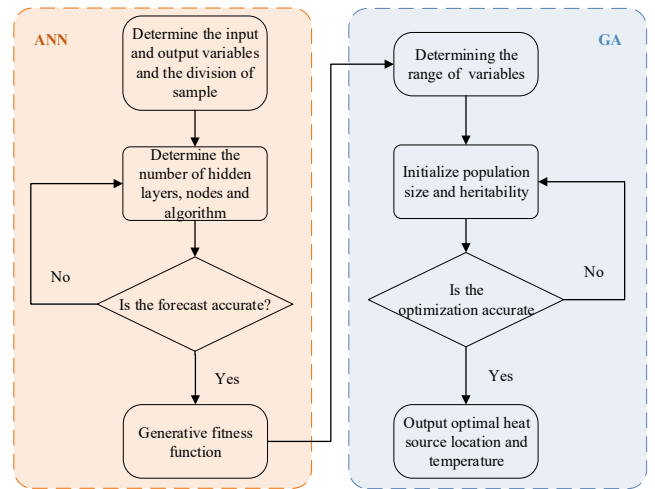
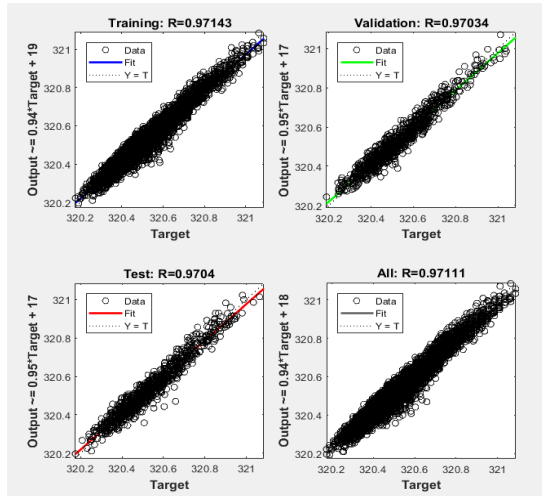


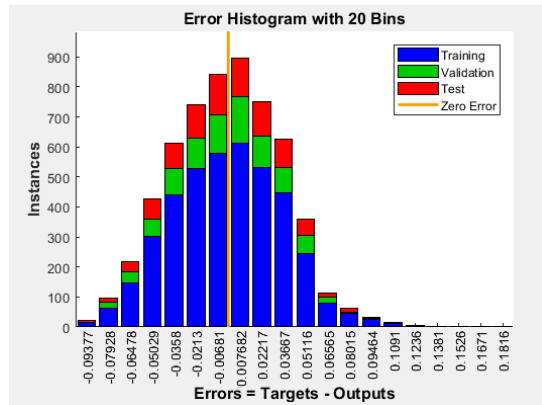
Fig. 7. GA-ANN optimization flow chart

III. RESULTS AND VALIDATION

The neural network was used to train the constrained samples, and the R coefficient and error histogram were selected as the evaluation indicators of the training performances. The closer the R value is to 1, the better the regression fit. The error histogram visually illustrates the deviation between predicted and true output values. Figures 8(a) and (b) depict the scatter plots and error histograms at each stage of the neural network, considering the highest temperature and average temperature weights as 0.2 and 0.8, respectively. Notably, the R value surpasses 0.97, indicating a high level of accuracy in the neural network fitting. Furthermore, the error distribution is predominantly concentrated near the zero line, with an error zero line of 0.0067. These findings further establish the precision and effectiveness of the neural network model. Subsequently, the trained neural network is utilized as the fitness function within the genetic algorithm optimization process to determine the optimal heat source position. Figure 9 illustrates the obtained optimal heat source position, wherein the optimization solution yields a maximum temperature of 320.182 K. From the results, it can be seen that the optimal heat source position satisfies the constraint of non-coincident within the interlayer coincident layer. In addition, the optimal arrangement of heat sources within each layer of the irregularly structured 3D ICs is observed in the overlapping region between chip layers. This observation reinforces the notion that heat dissipation is enhanced within the interlayer coincident region containing TSVs.



(a)



(b)

Fig. 8. Each evaluation index in the neural network: (a) odd plot of each stage, (b) error histogram

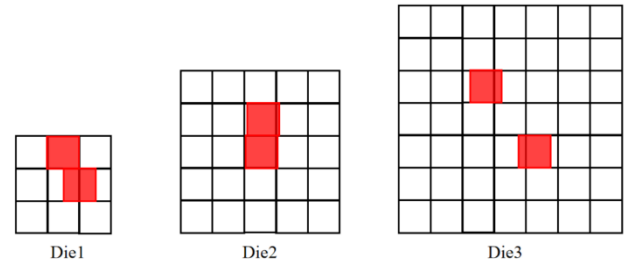
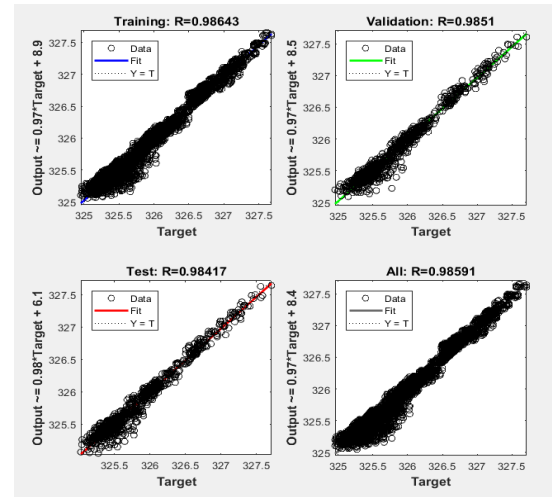
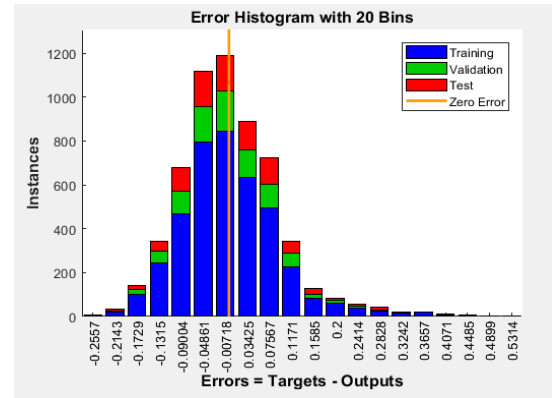


Fig. 9. The result map of the optimal position of the heat source

Similarly, the genetic algorithm and neural network underwent optimization, and the resulting scatter plots and error histograms at each stage of the neural network are displayed in Fig. 10. In this case, the highest temperature and average temperature weights were assigned values of 0.8 and 0.2, respectively. As can be seen from the figure, the R value exceeds 0.98, and the majority of errors cluster around the zero line. These findings underscore the high accuracy achieved by the neural network fitting. By employing the aforementioned weights, the optimization process yielded a maximum temperature of 324.638 K. The corresponding optimal position of the heat source is illustrated in Fig. 11. Notably, the optimal heat source arrangement successfully adheres to the constraint of non-coincidence within the interlayer coincident layer.



(a)



(b)

Fig. 10. Each evaluation index in the neural network: (a) odd plot of each stage, (b) error histogram

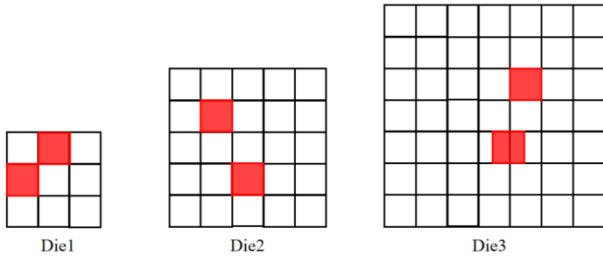


Fig. 11. The result map of the optimal position of the heat source

The optimal arrangement of heat source locations promotes efficient heat transfer, thereby reducing the maximum temperature in stacked chips. Simultaneously, this optimal arrangement ensures temperature uniformity within 3D ICs. The maximum temperature predictions and corresponding numerical simulation results for specific heat source locations are presented in TABLE III. The positions of the six heat sources are denoted by H1 to H6, respectively. Upon reviewing the table, it is evident that the predictions generated by the neural network closely align with the results obtained from numerical simulations. This convergence substantiates the high accuracy of the prediction model, providing a robust and precise prediction capability.

TABLE III. MAXIMUM TEMPERATURE PREDICTIONS AND NUMERICAL SIMULATIONS RESULTS AT SOME HEAT SOURCE LOCATIONS

H1	H2	H3	H4	H5	H6	Simulation values (K)	Prediction values(K)
7	1	3	8	4	2	327.142	327.423
5	3	8	1	2	7	327.543	327.334
6	6	2	2	7	1	332.377	332.355
4	4	8	1	6	3	332.600	332.038
1	7	3	6	7	2	327.082	327.236
2	7	3	1	6	5	327.215	327.650
4	7	7	6	3	2	329.031	328.728

To further validate the accuracy of the genetic algorithm optimization results, a comparison was conducted to determine the maximum temperature deviation between the optimization results and the numerical simulation results. The optimized maximum temperature value obtained through the genetic algorithm is 326.325 K. Figure 12 presents the numerical simulation results illustrating the optimal arrangement of heat sources when minimizing the maximum temperature. It is evident from the results that the maximum temperature achieved in the numerical simulation is 327.12 K, resulting in a deviation of merely 0.24% from the optimized maximum temperature of genetic algorithm. This negligible deviation serves as evidence of the high accuracy exhibited by the optimization method.

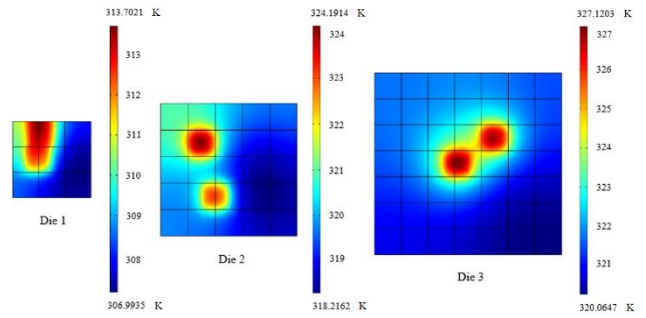


Fig. 12. Numerical simulation results of optimal heat source layout when maximum temperature is minimum

IV. CONCLUSION

This paper presents an effective methodology for characterizing and optimizing the heat source position in a three-layer irregularly structured 3D ICs. The proposed approach leverages a combination of genetic algorithm and artificial neural network techniques, while employing joint simulation with COMSOL and MATLAB and constrained sampling methods. Through this methodology, the optimal arrangement of heat source positions for each chip layer is obtained. To assess the accuracy and effectiveness of the proposed methodology, a comprehensive comparison is conducted between the optimization results and finite element simulation outcomes obtained using COMSOL. The results indicate a high level of consistency between the optimization results and the corresponding finite element simulation results.

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